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Environmental contamination and human health risk assessment of trace element exposure through diverse water sources

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Abstract

Human activities such as industrial effluents, agricultural run-off, and other land-based sources of pollution contaminate water with trace elements, posing threats to both the environment and human health. This study investigates concentrations of three trace elements, namely nickel (Ni), lead (Pb), and copper (Cu), in an assessment of health risks. The study comprised three different sources of water, namely surface water (SW), groundwater (GW), and industrial water (IW), sampled from six tehsils of District Bahawalpur, namely Bahawalpur, Bahawalpur Saddar, Hasilpur, Ahmedpur East, Yazman, and Khairpur Tamewali. A total of thirty water sampling sites, ten sites for each of the three types of water sources, were selected, and water samples were collected in the months of August, September, and October 2025. Results of the studied trace elements showed the highest concentrations in industrial water samples, followed by GW and SW samples ($p < 0.001$), with the highest concentration of Ni being 1.99 ± 0.20 mg/L, while that of Pb was 2.55 ± 0.11 mg/L and that of Cu was 3.24 ± 0.18 mg/L in IW samples. Significant differences were observed in the concentrations of the three studied trace elements in water samples as well as among the sampling sites of three types of water sources ($p < 0.001$). Estimated oral and dermal exposure (EE_{oral} and EE_{derm}) indicated low health exposure risks. However, THQ_{oral} values for Pb in GW and Ni, Pb and Cu in IW samples were found to exceed the acceptable threshold values. In addition to THQ_{oral} values, estimated cumulative non-carcinogenic health risks in the form of TTHQ_{oral} values for all three types of water sources under study were > 1 in all three water sources, suggesting potential risks. While, estimated carcinogenic health risks for the three types of water sources under study were found to be within the acceptable limits for all the studied health risks. This study provided a better understanding of various pollutants and their concentrations in water sources at the studied location.

Keywords: Water sources, Trace elements, Bioaccumulation, Pollution status, Health risk



1 Introduction

Water is an indispensable natural resource that sustains human health, ecosystem integrity, agricultural productivity, industrial development, and socioeconomic growth [11]. Although nearly 71% of the Earth's surface is covered by water, freshwater accounts for less than 3% of the global water reserves, and only a small fraction is readily accessible for human consumption and domestic use [64]. Rapid population growth, urbanization, industrial expansion, agricultural intensification, and climate change have substantially increased the demand for freshwater resources while simultaneously accelerating their deterioration in quality [64]. Consequently, ensuring access to safe drinking water has become one of the foremost environmental and public health challenges worldwide and remains a central objective of the United Nations Sustainable Development Goal 6 (SDG-6), which aims to ensure universal access to safe and clean water [28].

Among the numerous contaminants threatening freshwater resources, trace elements have received considerable scientific attention because of their persistence, non-biodegradability, environmental mobility, and potential to accumulate in aquatic ecosystems and living organisms. Unlike many organic pollutants that undergo degradation over time, trace elements remain in environmental compartments for prolonged periods and continuously cycle through water, sediments, soils, and biota. Their accumulation in drinking water has therefore become a major environmental concern owing to their adverse ecological impacts and their ability to cause long-term human exposure [3, 10], Vijay [63]. Increasing concentrations of trace elements in freshwater ecosystems have been associated with declining water quality, disruption of aquatic biodiversity, ecological imbalance, and growing risks to human health [41, 66].

Trace elements in aquatic environments are contributed by geogenic (natural) processes as well as by anthropogenic sources. Naturally, the metal-bearing rocks are weathered, minerals are dissolved, volcanic material is emitted, and soil is eroded to contribute to the surface water as well as to the groundwater. However, in recent decades, the anthropogenic sources have contributed to the majority of the contamination of the aquatic environments, i.e. industrial effluents, mining activities, municipal wastewater, agricultural runoff, fossil fuel combustion, metal processing industries, application of fertilizers and pesticides, landfill leachates, and rapid urbanization ecosystems [10, 52, 62]. These contaminants, introduced into the environment through various sources, are transported to different aquatic environments through runoff, atmospheric deposition, infiltration and groundwater recharge, thereby degrading both surface and groundwaters [62]. The untreated industrial and municipal wastewaters are continuously being discharged into the environment, and in developing countries, the lack of wastewater treatment facilities adds to the contamination in aquatic environments [39].

The contamination of trace elements in water is creating serious health problems as these elements can easily enter the human body through water. Some of the health effects caused by chronic exposure to trace elements are neurological problems, kidney problems, cardiovascular diseases, liver damage, skeletal problems, reproductive problems, developmental abnormalities and various types of cancer [16, 17], Vijay [63]. Therefore, to ensure the health of the people and to provide safe drinking water, guideline values for trace elements in drinking water have been established by international organizations such as the World Health Organization (WHO) [43].

Different trace elements have different biological functions and therefore they cannot be considered in a single group. While copper (Cu) is required for the human body in small quantities to carry out various physiological and biochemical activities like enzymatic activities, cellular respiration, antioxidant activities, formation of connective tissues, etc., high levels of this element can cause toxicity. It can cause disturbances in cellular activities, can cause increased oxidative stress, and can cause toxicity to the liver, kidneys, and gastrointestinal tract [19]. On the other hand, lead (Pb) is a non-essential element and has no known biological function in the human body. It is considered to be one of the most toxic elements found on the earth, and it causes cumulative toxicity to the human body. Even low levels of this element can cause toxicity to the human body after prolonged exposure. It can cause toxicity to the nervous system, blood, kidneys, heart, and reproductive organs [50]. While nickel (Ni) is naturally occurring and is widely used in various industrial applications, it is not required for the human body and can cause toxicity to the human body after prolonged exposure. It can cause various health problems like allergic skin reactions, respiratory problems, immunotoxicity, and carcinogenicity [16]. Therefore, it is very important to distinguish between essential and non-essential trace elements to find out their environmental behaviour, toxicological properties, and potential human health risks.

There is a constant struggle to provide safe drinking water in Pakistan as the country grapples to cope with the effects of rapid urbanization, industrial growth, lack of effective wastewater management, intensification of agricultural activities, and a surging population. Most of the country's drinking water is sourced from groundwater, which is being continuously threatened and has thus become a matter of national public health concern, requiring adequate protection [46, 49]. Water from different sources, including surface water, groundwater, wastewater, and industrial effluent, is being contaminated with industrial waste, agricultural runoff containing a mix of agricultural chemicals, sewage from cities and mines, and geological processes of the natural environment [21, 23]. Furthermore, the large volume of water being generated from seasonal flooding and uncontrolled urban growth is being freely discharged into all types of freshwater resources of the country and causing unprecedented water pollution, leading to widespread occurrence of water-borne diseases and serious chemical contamination [1]. National assessments conducted by the Pakistan Council of Research in Water Resources (PCRWR) have also demonstrated that a considerable proportion of drinking water sources fail to comply with recommended national and international water quality standards, highlighting the urgent need for continuous monitoring and effective management of drinking water resources [24].

Recent international studies have considerably advanced drinking water quality assessment by integrating trace element determination with pollution indices, multivariate statistical analyses, geospatial techniques, and quantitative human health risk assessment models. Such integrated approaches have enabled researchers to identify contamination sources, distinguish between geogenic and anthropogenic inputs, and evaluate both non-carcinogenic and carcinogenic risks associated with long-term exposure to contaminated drinking water. These comprehensive assessment frameworks provide a more robust understanding of the relationships between trace

element contamination and public health and have become increasingly important for developing evidence-based environmental management strategies [42, 53].

In Pakistan, numerous investigations have assessed drinking water quality and the occurrence of trace elements in different provinces, including Punjab, Sindh, Khyber Pakhtunkhwa, and Balochistan. These studies have consistently demonstrated that industrial effluents, agricultural runoff, municipal wastewater, mining activities, and natural geological processes contribute substantially to trace element contamination in both groundwater and surface water [12]. Several investigations have further evaluated the potential human health risks associated with contaminated drinking water using hazard quotient (HQ), hazard index (HI), and carcinogenic risk (CR) models. While these studies have significantly improved the understanding of drinking water contamination in Pakistan, many have been limited to specific geographical regions, individual water sources, or selected contaminants [13]. Moreover, most investigations have primarily focused on reporting contaminant concentrations and basic health risk indices, whereas comprehensive comparisons among multiple water sources using integrated health risk assessment frameworks remain relatively limited [39].

Research worldwide is utilizing trace element determination, water pollution assessment indices, water quality statistics analysis, GIS spatial analysis, and human health risk assessment, etc., to evaluate drinking water source contamination, its causes, geogenic vs. anthropogenic, water quality health risks, etc. [42, 56, 58]. Comprehensive studies on drinking water source contamination assessment utilizing the above approaches have not only supported scientists and decision-makers in better understanding the relationship between water contamination by trace elements and water-borne health risks but also have enabled them in planning, designing, and implementing effective environment-friendly health management measures.

Nevertheless, contamination characteristics vary considerably among different geographical regions because of differences in geology, land use, industrial development, agricultural practices, and population density. Consequently, region-specific investigations remain essential for generating reliable baseline data and supporting local water quality management [7]. Despite these important advances, significant knowledge gaps remain regarding the distribution of trace elements and their associated human health risks in several rapidly developing regions of Pakistan. Southern Punjab, despite experiencing increasing urbanization, agricultural intensification, industrial growth, and expanding municipal activities, has received comparatively less attention than many other parts of the country. Bahawalpur District relies predominantly on groundwater for domestic consumption, while surface water and industrial water also contribute to meeting municipal, agricultural, and industrial demands. However, comprehensive information regarding trace element contamination across these different water sources and its potential implications for human health remains limited. To the best of our knowledge, no previous study has systematically compared groundwater, surface water, and industrial water within Bahawalpur District while simultaneously evaluating trace element contamination together with quantitative human health risk assessment using internationally accepted risk assessment models. This represents an important knowledge gap that must be addressed to support evidence-based environmental management and public health protection in the region.

Copper (Cu), lead (Pb) and nickel (Ni) are among the most common trace elements, which have been reported in various studies conducted on drinking water quality around the world. These trace elements occur in the environment due to geogenic as well as anthropogenic sources, such as industrial effluents, metal processing units, municipal wastewater, agricultural activities, corrosion of water distribution pipes, fossil fuel combustion and weathering of metal bearing rocks [37]. These elements can enter drinking water sources from these sources and can be used as indicators of water quality with reference to environmental hazards. Concentrations of these trace elements in drinking water are reported in the form of guidelines by national as well as international organizations. Other trace elements such as cadmium (Cd), arsenic (As) and mercury (Hg) are also considered as very toxic elements but have not been included in the present study, as Cu, Pb and Ni are very common, persistent in environment, having pronounced adverse health effects in human being and have been reported in vast number of studies all over the world. Moreover, these three elements are included in the drinking water quality guidelines all over the world and, therefore, inclusion of these three elements in the present study would provide a solid base to correlate concentrations of these elements with contamination as well as health risks to the consumers of water under study.

Unlike many previous investigations conducted in Pakistan that evaluated either groundwater or surface water independently or primarily reported contaminant concentrations, the present study adopts a comprehensive approach by simultaneously assessing trace element, copper (Cu), lead (Pb), and nickel (Ni) concentrations in groundwater, surface water, and industrial water collected from six tehsils of Bahawalpur District. Furthermore, quantitative human health risk assessment models were employed to evaluate the potential non-carcinogenic and carcinogenic risks associated with oral and dermal exposure to these trace elements through drinking water consumption. The findings provide a comprehensive baseline dataset for Bahawalpur District and contribute valuable scientific evidence for improving drinking water quality management, protecting public health, and supporting the sustainable management of freshwater resources in southern Punjab, Pakistan.

2 Materials and methods

2.1 Study area

This study investigated the concentrations of nickel (Ni), lead (Pb), and copper (Cu) in water samples collected from three major water sources, namely surface water (SW), groundwater (GW), and industrial water (IW), and evaluated the associated human health risks. Water samples were collected from representative groundwater, surface water, and industrial water sources located across six tehsils of District Bahawalpur, Punjab, Pakistan, namely Bahawalpur City, Bahawalpur Saddar, Ahmadpur East, Yazman, Hasilpur, and Khairpur Tamewali. The geographic coordinates of all sampling locations were recorded using a Global Positioning System (GPS) to facilitate site identification, spatial visualization, and future monitoring (Table 1). The corresponding tehsil for each sampling location was identified using reverse geocoding based on the recorded geographic coordinates. In total, the 30 sampling sites were distributed among Bahawalpur City (n=6), Bahawalpur Saddar (n=6),

Table 1 Coordinates and of all sampling sites at each source of water

Source of water	Location #	Co-ordinates		Tehsil
		Latitudes °N	Longitudes °E	
Surface water	Site 1	29.40547	71.701342	Bahawalpur City
	Site 2	29.39153	71.804398	Bahawalpur Saddar
	Site 3	29.6387	72.377661	Khairpur Tamewali
	Site 4	29.79471	72.545914	Hasilpur
	Site 5	29.17196	71.263562	AhmadPur East
	Site 6	29.25998	71.060596	AhmadPur East
	Site 7	29.33045	71.562683	Bahawalpur City
	Site 8	29.30135	71.447428	Bahawalpur Saddar
	Site 9	29.10263	71.601759	Yazman
	Site 10	29.22147	71.818965	Yazman
Ground water	Site 1	29.36078	71.701374	Bahawalpur Saddar
	Site 2	29.37412	71.608614	Bahawalpur City
	Site 3	29.31351	71.536108	Bahawalpur Saddar
	Site 4	29.32189	71.566425	Bahawalpur City
	Site 5	29.65113	72.507186	HasilPur
	Site 6	29.11892	71.291215	AhmadPur East
	Site 7	29.21199	71.040092	AhmadPur East
	Site 8	29.12134	71.753721	Yazman
	Site 9	29.15273	71.776961	Yazman
	Site 10	29.55344	72.200314	Khair Pur Tamewali
Industrial water	Site 1	29.4039	71.658582	Bahawalpur City
	Site 2	29.33702	71.568681	Bahawalpur Saddar
	Site 3	29.73638	72.549161	HasilPur
	Site 4	29.58561	72.233716	Khairpur Tamewali
	Site 5	29.33095	71.586484	Bahawalpur City
	Site 6	29.29774	71.520263	Bahawalpur Saddar
	Site 7	29.09001	71.716351	Yazman
	Site 8	29.17511	71.731598	Yazman
	Site 9	29.10521	71.00474	AhmadPur East
	Site 10	29.17221	71.257608	AhmadPur East

Ahmadpur East (n = 6), Yazman (n = 6), Hasilpur (n = 3), and Khairpur Tamewali (n = 3). The study area map (Fig. 1) was prepared using QGIS version 4.2.1 (Belém do Pará). Administrative boundary datasets at the provincial, district, and tehsil levels were obtained from the Humanitarian Data Exchange (HDX) database (<https://data.humdata.org/>), while the hydrographic network (major rivers and canals) was incorporated using publicly available HDX waterway datasets. The georeferenced sampling locations were overlaid on the administrative boundary and hydrographic layers, and distinct symbols and colors were used to differentiate surface water, groundwater, and industrial water sampling sites. The final figure layout, including map composition, cartographic elements, annotations, and graphical refinements, was completed using Inkscape version 1.4.2 (f4327f4, 2025-05-13). Ten sampling sites were selected for each water source (SW, GW, and IW), and three replicate samples were collected from each site. Sampling was conducted between August and October 2025.

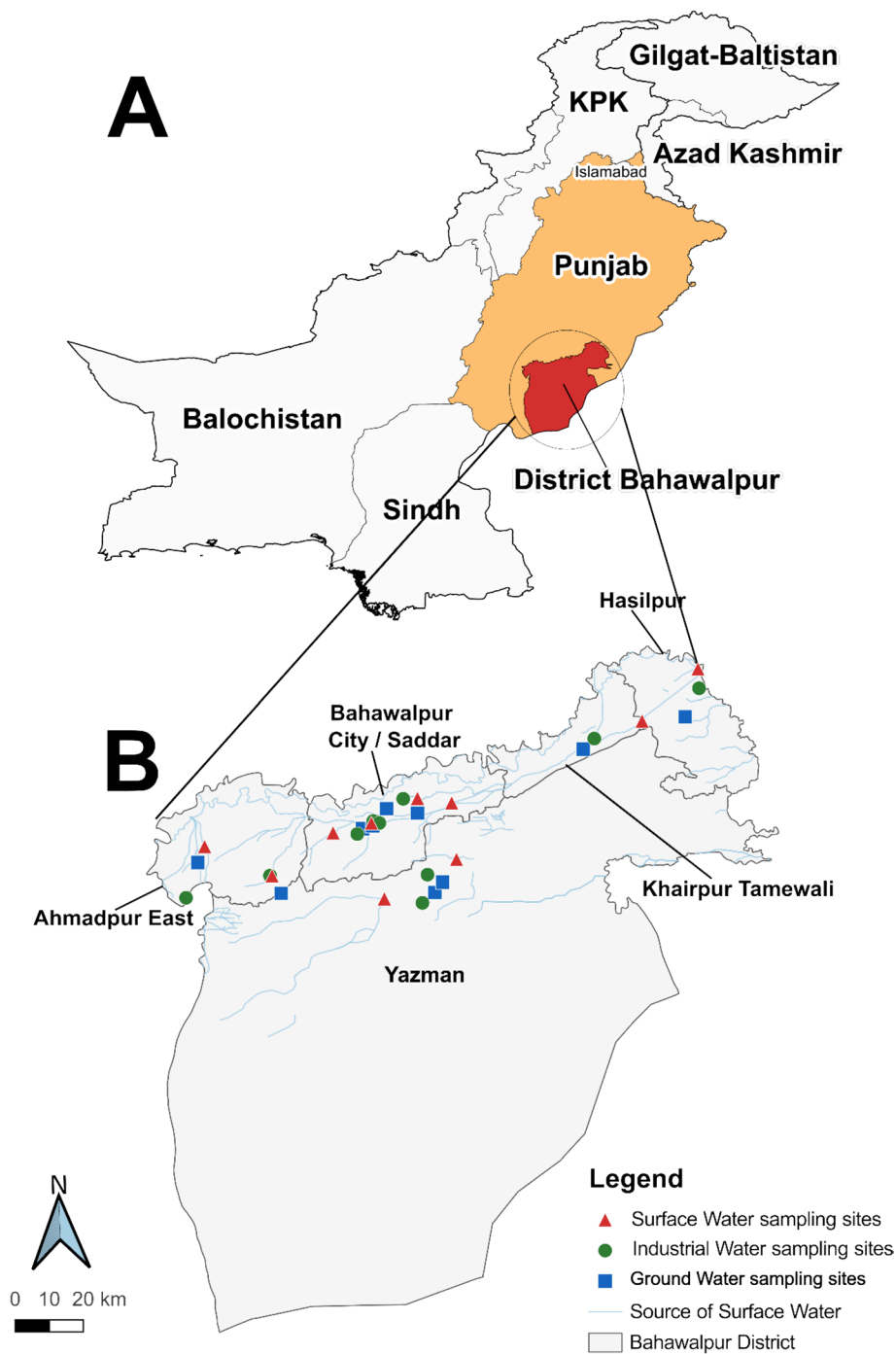


Fig. 1 Location of the study area and distribution of sampling sites in southern Punjab, Pakistan. Location of Bahawalpur District within Punjab Province, Pakistan. Distribution of surface water, groundwater, and industrial water sampling sites across the study area showing tehsil boundaries and the major hydrographic network

2.2 Collection, transportation and storage of water samples

A representative sampling approach was adopted to collect water samples from three different water sources. Groundwater, surface water, and industrial water were selected as representative water sources because groundwater is extensively used for drinking purposes, surface water supports aquatic ecosystems and is widely used for domestic and livestock activities, while industrial effluents discharged into drainage channels ultimately enter freshwater systems and constitute a major source of trace element contamination. For each water source, ten sampling points were selected from different locations across the study area based on their accessibility and suitability for sample collection and transportation while ensuring representative coverage of the respective water source. Groundwater sampling points were selected from water pumps located within mosque premises because these water sources are routinely used by local communities for drinking and ablution purposes. Sampling points of surface water sources were selected from major canals in different geographical locations of the study area. Sampling points of industrial water sources were chosen from discharge points of active sugar mills, textile mills, flour mills, and oil mills. Industrial water samples were collected at the discharge outlets in the drain or the water body. This was done right after the water had been discharged into the drain or water body. The surface water and industrial water samples were collected at a depth of approximately 150 cm below water surface to minimize the influence of surface debris and floating contaminants.

The collected samples of water were stored in pre-cleaned high-density polyethylene (HDPE) bottles. Before the samples were collected, the HDPE bottles were cleaned with laboratory detergent and then rinsed with tap water, immersed in 5% (v/v) nitric acid solution for more than 24 h, and rinsed again several times with double-distilled deionized water. Each sample bottle was rinsed two to three times with water from the sampling site immediately prior to sampling in order to condition the bottle without diluting the sample because there might be some deionized water left in the bottle after cleaning. Field rinsing was done purely for conditioning purposes and not for the purpose of changing the concentration of trace elements. After collection of samples, bottles were closed, placed in a portable insulated ice box containing ice packs, and maintained at temperatures 6 ± 3 °C using ice packs during transportation. Samples were stored at appropriate conditions in the Zoology Laboratory, Government Sadiq College Women University, Bahawalpur for subsequent analysis.

2.3 Sample processing and digestion

A total of 90 water samples (30 samples from each water source) were collected from the selected sampling sites. These samples were analyzed for the concentration of Ni, Pb, and Cu to carry out human health risk assessment.

50 ml of each water sample was placed in a beaker and left to cool down. 1 ml of concentrated nitric acid (HNO₃; 69–72%) and 2.5 ml of concentrated hydrochloric acid (HCl; 30%) were added to the beaker, and the beaker was placed on a hot plate (Westpoint Double Hot Plate WF-262). The mixture was then heated to 90 °C and left to mix until the volume of the mixture had decreased to approximately 20 ml.

After cooling, the solutions were filtered through Whatman No. 42 filter paper in a 100 mL volumetric flask. The solutions were then diluted to volume with deionized water. 2 mL of concentrated HNO_3 was added to the solutions to ensure complete acidification and to obtain clear solutions. The solutions were then diluted with 3% (v/v) nitric acid according to the USEPA digestion protocol. After filtration and settling, the solutions were transferred into pre-cleaned sample containers which had been sterilized at 135°C . The samples were then sealed properly and stored until the start of the trace element analysis. For trace element analysis of digested samples Atomic Absorption Spectrophotometer (AAS), Analytik Jena NovAA 400P, at Central Laboratory, Muhammad Nawaz Sharif University of Agriculture, Multan, was used.

2.4 Determination of trace elements

Prior to analysis, the Atomic Absorption Spectrophotometer (AJ NovAA 400P, Analytik Jena, Germany) was optimized under the manufacturer's recommended flame conditions for the determination of Ni, Pb, and Cu. Double-distilled and deionized water was used to prepare all working solutions and served as the reagent blank. Calibration standards of 0, 1, 2, 4, 6, and 8 mg L^{-1} were prepared from certified stock standard solutions. Calibration curves were generated by plotting absorbance against the corresponding standard concentrations and fitting the data using linear least-squares regression ($y = a + bx$). Blank correction was applied prior to the quantification of sample concentrations. The concentrations of Ni, Pb, and Cu were determined according to the *Standard Methods for the Examination of Water and Wastewater* [4], Methods 3500-Ni B, 3500-Pb B, and 3500-Cu B). The trace element analyses were performed at an external analytical laboratory following the standard analytical protocol. Three independent water samples collected from each sampling location were analysed separately. One concentration value was obtained from each independently collected sample using AAS, and the results are presented as mean $\pm$ standard deviation (SD) of the three independent samples.

2.5 Human health risk assessment

This study estimated two potential toxicity parameters concerning human health, which include Oral toxicity due to ingestion and dermal toxicity caused by the contact of human skin with toxic trace elements. The equations below were used to estimate the potential exposure of trace elements according to [14, 15, 25]:

$$EE_{oral} = \frac{C_i \times R_i \times E_f \times E_d}{BW \times AD} \quad (1)$$

$$EE_{derm} = \frac{C_i \times E_f \times E_d \times C_f \times E_{td} \times E_{ss} \times K_{dp}}{BW \times AD} \quad (2)$$

where EE_{oral} is the ingestion exposure from water; C_i shows the observed concentration in water; R_i is the ingestion rate (2 l/day); E_f is the exposure frequency in days (365 days per year); E_d represents the exposure duration in years (30 years); BW is the average body weight (70 kg); AD represents average days of exposure (10,950 days); C_f is the unit conversion factor (10^{-6} kg per milligram); E_{td} is the exposure time per day (2 h/day);

E_{SS} is the Exposed surface area of Skin (5700 cm^2) and K_{dp} is the dermal permeability co-efficient (0.001 cm/h).

Furthermore, target hazardous quotients (THQs) and total target hazardous quotients (TTHQs) were used to estimate the potential non-carcinogenic health effects from exposure (oral and dermal) to trace elements (Ni, Cu) according to [59] health risk detection guidelines. THQ and TTHQ were calculated only for Ni and Cu because no oral RfD was used for Pb, while carcinogenic risk (CR) was calculated only for Pb using the cancer slope factor (CSF). The equations for THQ and $TTHQ$ are given below (Eqs. 3 and 4).

$$THQ_{oral/derm} = \frac{EE_{oral/derm}}{RfD} \quad (3)$$

$$TTHQ_{oral/derm} = \sum_{i=1}^n HQ_{oral/derm} \quad (4)$$

Here, $THQ_{oral/derm}$ is the target hazardous quotient for via oral intake or dermal contact under the respective exposure amount; RfD represents the reference dose for the contaminant causing hazardous effect. For the current study, the RfD (mg/kg/day) values of the studied trace elements are 0.02 and 0.04 for Ni and Cu respectively. RfD value for Pb is not established, hence THQ for Pb was not calculated. $TTHQ_{oral/derm}$ total target hazardous quotient is the sum of individual $THQ_{oral/derm}$ of each trace element. The RfD value for trace elements is considered to be the same for both dermal and oral routes [25].

2.6 Carcinogenic risk factor (CR)

The carcinogenic effect of oral and dermal contact of the studied trace elements factor was calculated using Eq. 5 [29]:

$$CR_{oral/derm} = EE_{\frac{oral}{derm}} \times CS_f \quad (5)$$

where CS_f is the carcinogenic Slope factor (mg/kg/day). It is 0.0085 for Pb [5, 60], however, this has not been calculated for Ni and Cu.

2.7 Statistical analysis

All statistical analyses were done by IBM SPSS Statistics 25.0 software (IBM Corp., Armonk, NY, USA). The concentration of Ni, Pb, and Cu was presented as mean $\pm$ SD (Standard Deviation) based on three replicates for each sampling location. Before conducting the ANOVA, the assumptions of normal distribution and homogeneity of variance were checked with the Shapiro–Wilk test and Levene’s test, respectively. Two-way ANOVA was carried out separately for each water sample (Surface water, Ground water, and Industrial water) to examine the influence of trace metal type, sampling locations, and their interaction on trace element concentrations. If there was any significant effect was observed ($P < 0.05$), Tukey’s Honestly Significant Difference (HSD) test was used to conduct multiple comparisons between sampling locations and trace metals. The level of significance was considered as $P < 0.05$. Mean values bearing the

same letter(s) did not differ significantly, but the means with different letters differed significantly according to Tukey's HSD test.

3 Results

The concentrations of Ni, Pb, and Cu in surface water (SW), groundwater (GW), and industrial water (IW) at ten sampling sites in District Bahawalpur, Punjab, Pakistan are presented in Table 2. Spatial variation in trace element concentrations at the sampling sites within the three water types was observed.

The range of concentrations for Ni, Pb, and Cu in surface water samples is shown in Table 2. The highest concentration of Ni was found at Site 5, with a concentration of 0.43 ± 0.02 mg/L, whereas Sites 6 and 10 had the lowest concentrations with values of 0.08 ± 0.02 mg/L. For Pb, the highest concentration was found at Site 3 with a concentration of 0.44 ± 0.01 mg/L. The lowest concentration of Pb was found at Site 5. For Cu, the highest concentration was found at Site 10 with a concentration of 0.63 ± 0.04 mg/L, whereas the lowest concentration of Cu was found at Site 4 with a concentration of 0.45 ± 0.03 mg/L.

Concentrations of trace elements in groundwater at the study area showed considerable spatial variations. Nickel concentrations in the groundwater samples ranged from 0.38 ± 0.02 mg/L to 0.48 ± 0.03 mg/L. The highest concentration of nickel was found in the sample collected from Site 8, whereas the lowest concentration was found in the sample collected from Site 7. Lead concentrations in the groundwater samples ranged from 0.68 ± 0.05 mg/L to 0.75 ± 0.06 mg/L. The highest concentration of lead in the groundwater samples was found to be 0.75 ± 0.06 mg/L at Sites 9 and 4. The samples from Sites 1, 4 and 9 contained 0.74 ± 0.03 mg/L, 0.75 ± 0.04 mg/L and 0.75 ± 0.06 mg/L of lead, respectively. The lowest concentration of lead in the groundwater samples was found to be 0.68 ± 0.05 mg/L at Sites 2 and 8. Copper concentrations in the groundwater samples ranged from 0.86 ± 0.04 mg/L to 0.96 ± 0.08 mg/L. The highest concentration of copper was found to be 0.96 ± 0.08 mg/L at Site 8. The lowest concentration of copper was found to be 0.86 ± 0.04 mg/L at Site 5.

A very large number of samples from industry contained high concentrations of the three investigated trace elements. The concentration of the element nickel in all samples, which were analyzed for this element, ranged from 1.18 ± 0.13 mg/L (Site 1) to 1.99 ± 0.20 mg/L (Site 7). The concentration of lead in all samples, which were analyzed for this element, ranged from 1.89 ± 0.06 mg/L (Site 3) to 2.55 ± 0.11 mg/L (Site 9). The highest concentration of the element copper in all analyzed samples was 3.24 ± 0.18 mg/L (Site 8). The lowest concentration was found in samples from Site 4 with 2.34 ± 0.07 mg/L.

A comparison of the concentrations of Ni, Pb and Cu in the three water types showed distinct distribution patterns. The highest concentrations of all three trace elements were found in the industrial water, followed by groundwater and surface water. In industrial water as well as in the groundwater, Cu was the most abundant trace element, followed by Pb and Ni. In the surface water, however, the distribution pattern was different from the two other water types, with the highest concentrations of Pb, followed by Cu and Ni.

The highest concentration of trace elements was found for Cu in industrial water at Site 8 (3.24 ± 0.18 mg/L). The lowest concentration of Ni was found in surface water at Sites

Table 2 Concentration (mg/L $\pm$ S.D) of trace elements (Ni, Pb, and Cu) at different sites of three water sources (surface, ground, and industry) from the district of Bahawalpur, Punjab, Pakistan

Source of water	Trace metal	Sampling locations									
		Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8	Site 9	Site 10
Surface	Ni	0.09 $\pm$ 0.02 ^{fz}	0.15 $\pm$ 0.02 ^{ez}	0.35 $\pm$ 0.01 ^{bz}	0.21 $\pm$ 0.01 ^{dz}	0.43 $\pm$ 0.02 ^{Az}	0.08 $\pm$ 0.03 ^{fz}	0.35 $\pm$ 0.02 ^{bz}	0.11 $\pm$ 0.01 ^{efz}	0.26 $\pm$ 0.03 ^{cz}	0.08 $\pm$ 0.02 ^{fz}
	Pb	0.41 $\pm$ 0.02 ^{cy}	0.38 $\pm$ 0.02 ^{ey}	0.44 $\pm$ 0.01 ^{ay}	0.39 $\pm$ 0.03 ^{dy}	0.33 $\pm$ 0.05 ^{hy}	0.40 $\pm$ 0.02 ^{cy}	0.43 $\pm$ 0.02 ^{by}	0.38 $\pm$ 0.03 ^{ey}	0.34 $\pm$ 0.04 ^{by}	0.36 $\pm$ 0.05 ^{fy}
	Cu	0.56 $\pm$ 0.03 ^{dx}	0.48 $\pm$ 0.04 ^{fx}	0.53 $\pm$ 0.02 ^{ex}	0.45 $\pm$ 0.03 ^{gx}	0.58 $\pm$ 0.02 ^{cx}	0.61 $\pm$ 0.01 ^{bx}	0.59 $\pm$ 0.03 ^{cx}	0.49 $\pm$ 0.02 ^{fx}	0.53 $\pm$ 0.02 ^{ex}	0.63 $\pm$ 0.04 ^{ax}
Ground	Ni	0.44 $\pm$ 0.02 ^{dz}	0.39 $\pm$ 0.04 ^{hz}	0.41 $\pm$ 0.02 ^{fz}	0.47 $\pm$ 0.05 ^{bz}	0.45 $\pm$ 0.05 ^{cz}	0.40 $\pm$ 0.02 ^{gz}	0.38 $\pm$ 0.02 ^{hz}	0.48 $\pm$ 0.03 ^{az}	0.43 $\pm$ 0.03 ^{ez}	0.46 $\pm$ 0.02 ^{bz}
	Pb	0.74 $\pm$ 0.03 ^{ay}	0.68 $\pm$ 0.05 ^{fy}	0.70 $\pm$ 0.04 ^{dy}	0.75 $\pm$ 0.04 ^{ay}	0.73 $\pm$ 0.06 ^{by}	0.69 $\pm$ 0.04 ^{ey}	0.72 $\pm$ 0.05 ^{cy}	0.68 $\pm$ 0.05 ^{fy}	0.75 $\pm$ 0.06 ^{ay}	0.71 $\pm$ 0.02 ^{cy}
	Cu	0.91 $\pm$ 0.07 ^{dx}	0.88 $\pm$ 0.08 ^{ex}	0.92 $\pm$ 0.10 ^{cx}	0.91 $\pm$ 0.06 ^{dx}	0.86 $\pm$ 0.04 ^{fx}	0.94 $\pm$ 0.11 ^{bx}	0.92 $\pm$ 0.09 ^{cx}	0.96 $\pm$ 0.08 ^{ax}	0.94 $\pm$ 0.10 ^{bx}	0.89 $\pm$ 0.04 ^{ex}
Industry	Ni	1.18 $\pm$ 0.13 ^{dz}	1.41 $\pm$ 0.14 ^{dz}	1.54 $\pm$ 0.16 ^{bcdy}	1.61 $\pm$ 0.11 ^{abcdy}	1.32 $\pm$ 0.11 ^{cdy}	1.68 $\pm$ 0.20 ^{abcy}	1.99 $\pm$ 0.20 ^{az}	1.34 $\pm$ 0.07 ^{cdz}	1.57 $\pm$ 0.15 ^{abcdy}	1.90 $\pm$ 0.16 ^{aby}
	Pb	2.12 $\pm$ 0.17 ^{bcdy}	2.14 $\pm$ 0.04 ^{bcdy}	1.89 $\pm$ 0.06 ^{dy}	2.19 $\pm$ 0.12 ^{bcdx}	2.32 $\pm$ 0.14 ^{abx}	1.97 $\pm$ 0.08 ^{cdy}	2.36 $\pm$ 0.07 ^{aby}	2.09 $\pm$ 0.05 ^{bcdy}	2.55 $\pm$ 0.11 ^{ax}	2.34 $\pm$ 0.05 ^{abx}
	Cu	2.52 $\pm$ 0.07 ^{dex}	2.74 $\pm$ 0.08 ^{bcdx}	2.64 $\pm$ 0.21 ^{cdex}	2.34 $\pm$ 0.07 ^{ex}	2.57 $\pm$ 0.06 ^{dex}	2.99 $\pm$ 0.11 ^{abcx}	3.06 $\pm$ 0.10 ^{abx}	3.24 $\pm$ 0.18 ^{ax}	2.81 $\pm$ 0.16 ^{bcdx}	2.70 $\pm$ 0.18 ^{bcdx}

Values are expressed as mean $\pm$ SD. The means labelled with different small letters (a–h) in the same row are significantly different among sampling sites based on Tukey's HSD test ($P < 0.05$). The means with different capital letters (X–Z) in the same column in each water source are significantly different among the trace metals (Ni, Pb and Cu) using Tukey's HSD test ($P < 0.05$)

6 and 10 (0.08 ± 0.03 mg/L and 0.08 ± 0.02 mg/L). Highest concentrations for Cu were found in groundwater at all sites, with a maximum concentration of 0.96 ± 0.08 mg/L for Site 8. The lowest concentration of Ni was found at Site 7 with 0.38 ± 0.02 mg/L. This underlines large differences in the distribution of trace elements within the investigated water types. While large amounts of trace elements are found in industrial water, their amount in groundwater and surface water are far below those in industrial water.

A two-way ANOVA test was conducted for the concentrations of trace elements in three types of water (surface water, underground water, and rainwater), and the results are presented in Table 3. First, the test for surface water showed that trace element concentration was significantly affected by trace element type ($df=2$, $F=10,776.063$, $p<0.001$), water sampling location ($df=9$, $F=264.304$, $p<0.001$), and the interaction of the two ($df=18$, $F=278.611$, $p<0.001$). Also, the two-way ANOVA for the groundwater samples showed significant effects of the trace element type ($df=2$, $F=191,198.051$, $p<0.001$), of the sampling location ($df=9$, $F=343.984$, $p<0.001$), and of the interaction of both factors ($df=18$, $F=278.211$, $p<0.001$).

The concentrations of trace elements in industrial water were also significantly affected by trace element type ($df=2$, $F=624.667$, $p<0.001$), by sampling location ($df=9$, $F=13.714$, $p<0.001$) and by the interaction between both factors ($df=18$, $F=8.352$, $p<0.001$). Thus, trace element type and sampling location affected the trace element concentrations in all the investigated water sources.

To calculate the non-cancer health hazards, risk assessments on human health were carried out to evaluate the possible health hazards related to Ni, Pb, and Cu exposure through ingestion and dermal contact via surface water (SW), groundwater (GW), and industrial water (IW). Exposure estimates (EE_{oral} and EE_{derm}), target hazard quotients (THQ_{oral} and THQ_{derm}), total target hazard quotients (TTHQ_{oral} and TTHQ_{derm}), and carcinogenic risks (CR_{oral} and CR_{derm}) (Table 4) were used for evaluating health risks (non-cancer and cancer). When considering the surface water samples,

Table 3 Two-way analysis of variance (ANOVA) for type of metal, sampling location and interaction between them for each source of water

Source of water	Source of variance	Df	Mean square	F	p
Surface	Metal	2	.842	10,776.063	<0.001
	Location	9	.021	264.304	<0.001
	Metal $\times$ Location	18	.022	278.611	<0.001
	Error	60	7.810E-5		
	Total	89			
Ground	Metal	2	1.722	191,198.051	<0.001
	Location	9	.003	343.984	<0.001
	Metal $\times$ Location	18	.003	278.211	<0.001
	Error	60	9.006E-6		
	Total	89			
Industrial	Metal	2	10.959	624.667	<0.001
	Location	9	.241	13.714	<0.001
	Metal $\times$ Location	18	.147	8.352	<0.001
	Error	60	.018		
	Total	89			

Table 4 Human health risk assessment of water sampled from various sources

Source of water	Metal	$EE_{oral} \times 10^{-2}$	$EE_{derm} \times 10^{-8}$	THQ_{oral}	$THQ_{derm} \times 10^{-6}$	$TTHQ_{oral}$	$TTHQ_{derm} \times 10^{-6}$	$CR_{oral} \times 10^{-4}$	$CR_{derm} \times 10^{-9}$
Surface	Ni	0.60	3.44	0.30	1.72	0.69 [†]	3.94	–	–
	Pb	1.11	6.29	– [*]	–			0.94	0.54
	Cu	1.57	8.88	0.39	2.22			–	–
Ground	Ni	1.23	7.02	0.62	3.51	1.27	7.23	–	–
	Pb	2.04	11.64	–	–			1.73	0.99
	Cu	2.61	14.87	0.65	3.72			–	–
Industrial	Ni	4.44	25.31	2.22	12.66	4.19	23.90	–	–
	Pb	6.28	35.78	–	–			5.34	3.04
	Cu	7.89	44.96	1.97	11.24			–	–

^{*} THQ values for Pb were not calculated because no scientifically justified oral reference dose (RfD) for inorganic Pb is currently available under the USEPA IRIS human health risk assessment framework

[†] TTHQ represents the total target hazard quotient and was calculated as the sum of the individual THQ values for Ni and Cu only (Pb excluded)

EE: estimated exposure; THQ: target hazard quotient; TTHQ: total target hazard quotient; CR: carcinogenic risk; RfD: reference dose

the individual THQ for the oral ingestion of Ni (0.30) and Cu (0.39) was less than the threshold value of 1, which means that there is no risk associated with the individual trace elements in terms of non-carcinogenic effect. But even TTHQ_{oral} (total hazard quotient for oral intake) value (0.69) is less than the threshold value, which implies that combined ingestion of Ni and Cu does not result in any adverse non-carcinogenic effect. The THQ values for Ni (1.72×10^{-6}) and Cu (2.22×10^{-6}), as well as TTHQ_{derm} (cumulative risk) (3.94×10^{-6}), for dermal exposure are very low. The estimated carcinogenic risks through oral (0.94×10^{-4}) and dermal (0.54×10^{-9}) routes of exposure for Pb are also low. The groundwater had higher health risk index values than surface water. While the individual THQs for oral exposure to nickel (Ni) and copper (Cu), 0.62 and 0.65, respectively, are still lower than the critical value of 1, the cumulative oral risk (TTHQ_{oral}=1.27) exceeds the limit, which suggests a possibility of non-carcinogenic health risk due to combined oral exposure to Ni and Cu. In the same way, the TTHQ_{derm} for dermal risk was still extremely low (7.23×10^{-6}). The carcinogenic risks due to Pb exposure, orally (1.73×10^{-4}) and dermally (0.99×10^{-9}) were estimated. Industrial water was the one that had the greatest potential to pose health hazards among all the water samples studied. Oral THQs for Ni (2.22) and Cu (1.97) values were above the threshold limit value of 1, while the overall oral TTHQ value of 4.19 was well above the threshold limit value of 1, indicating considerable potential to cause non-cancerous health problems due to oral intake of the industrial water sample. Dermal THQs for Ni (12.66×10^{-6}) and Cu (11.24×10^{-6}) values were higher than the corresponding THQs of surface and ground waters, but several orders of magnitude lower than 1. Carcinogenic risks for Pb were the highest in industrial water, where CR_{oral} value is 5.34×10^{-4} and CR_{derm} value is 3.04×10^{-9} .

A comparison was also made of the concentrations of Ni, Pb and Cu in the water samples from the three water sources and the drinking water guideline values for these elements as specified by the World Health Organization [65] and the United States Environmental Protection Agency [61] (Table 5). The results showed that the

Table 5 Comparison of trace element concentrations in water samples with drinking water guideline values established by WHO and USEPA

Source of water	Metal	Concentrations (mg/L) of trace element			Drinking water upper limits (mg/L)	
		Mean	Minimum	Maximum	[61]	[65]
Surface	Ni	0.21	0.08	0.43	–	0.07
	Pb	0.39	0.33	0.44	0.01	0.01
	Cu	0.54	0.45	0.63	1.30	2.0
Ground	Ni	0.43	0.38	0.48	–	0.07
	Pb	0.72	0.68	0.75	0.01	0.01
	Cu	0.91	0.86	0.96	1.30	2.00
Industrial	Ni	1.55	1.18	1.99	–	0.07
	Pb	2.20	1.89	2.55	0.01	0.01
	Cu	2.76	2.34	3.24	1.30	2.00

concentrations of Ni in the surface water, groundwater, and industrial water samples all exceeded the WHO drinking water guideline value for this element of 0.07 mg/L. The federal drinking water standard for Ni has not been established by the USEPA. The concentrations of Pb in all three of the water samples exceeded the USEPA's action level for this element of 0.015 mg/L and the WHO's drinking water guideline value for this element of 0.01 mg/L. The concentrations of Cu in the surface water and groundwater samples were all below the drinking water guideline values for this element specified by the WHO (2.0 mg/L) and the USEPA (1.3 mg/L), whereas the concentrations of Cu in the industrial water samples all exceeded these permissible limits for Cu in drinking water (Table 5).

The overall assessment of water samples indicates that all three samples require monitoring and management of their trace element concentrations as their levels have exceeded drinking water guideline values for one or more elements.

4 Discussion

In the present study, the concentration of Ni, Pb and Cu in surface water, groundwater and industrial water from six different locations in District Bahawalpur, Punjab, Pakistan was assessed, and their potential health risk to humans was determined. Our results showed a significant difference in the concentration of trace elements in various samples of water, and the highest concentration of trace elements was found in industrial water, followed by groundwater and surface water. Among the three trace elements, the highest concentration of Cu was found in industrial water and groundwater samples, followed by Pb in surface water samples. The highest concentration of Pb was found in surface water samples.

There is abundant evidence from Pakistan to support the fact that high concentrations of trace elements in groundwater and industrial water are because of high levels of industrial effluents, municipal wastewater, agricultural runoff, and natural geochemical processes [6, 32]. In developing countries like Pakistan, groundwater is the main source of drinking water for the majority of the population, whereas surface water is used for various purposes including domestic use, agriculture, and irrigation [38, 51]. Therefore,

degradation of these water sources is an issue of great environmental and health concern, as long-term exposure to high concentrations of trace elements could lead to health problems. Relatively low concentrations of trace elements in surface water may be due to dilution, high flow rates and dispersion of contaminants, whereas high concentrations in groundwater and industrial water may be due to accumulation of contaminants over long periods, limited natural dilution, and continued input of contaminants from anthropogenic activities [40].

The comparatively lower concentrations observed in surface water may be attributed to continuous water movement, dilution by rainfall and canal inflows, sedimentation processes, and shorter residence times, all of which reduce dissolved trace element concentrations. Nevertheless, surface water remains vulnerable to contamination through agricultural runoff, municipal wastewater discharge, atmospheric deposition, and urban stormwater, particularly during periods of intensive irrigation and seasonal rainfall [52]. These findings align with prior research that has reported similar results in various districts and regions across Pakistan. Numerous previous studies conducted within the country have also revealed the presence of high concentrations of dangerous metals in groundwater that is typically used for drinking [34, 36, 48].

Groundwater contamination is likely associated with the gradual migration of dissolved trace elements through soil profiles and aquifer materials. Leaching from contaminated surface environments, infiltration of agricultural runoff, seepage from industrial disposal sites, leakage from domestic wastewater systems, and weathering of local geological formations may collectively contribute to groundwater contamination. Because groundwater generally has a longer residence time than surface water, dissolved trace elements may accumulate over extended periods, resulting in comparatively higher concentrations. Furthermore, hydrogeological characteristics, including aquifer composition, soil permeability, groundwater flow dynamics, and recharge conditions, strongly influence the transport and persistence of trace elements within groundwater systems [62].

Agricultural activities also represent an important non-point source of trace element contamination within the study area. Long-term application of phosphate fertilizers, pesticides, livestock manure, and irrigation return flows may gradually enrich agricultural soils with trace elements that are subsequently transported into nearby surface waters through runoff or infiltrate groundwater through percolation. Therefore, the contamination observed in the present study is likely the combined result of industrial emissions, agricultural practices, domestic wastewater discharge, and natural geological processes rather than a single pollution source [10, 52, 55].

Environmental contamination by heavy metals is considered a major environmental problem, particularly in aquatic environments, due to their toxic nature. Moreover, the toxic effects of heavy metals increase with the presence of other heavy metals [33, 39]. In the present study, the atomic absorption spectrophotometric method was used to determine the concentration of heavy metals including Chromium (Cr), Iron (Fe), Nickel (Ni), Cadmium (Cd), Lead (Pb), Zinc (Zn), Cobalt (Co) and Manganese (Mn) in the water samples collected from the groundwater of industrial area of Kasur, Pakistan. The concentrations of these heavy metals were found to be higher than the water quality standards for these parameters as specified by World Health Organization

(WHO). The concentrations of heavy metals in the water samples of Kasur industrial area were found to be in the following order: Chromium (Cr) > Lead (Pb) > Iron (Fe) > Cadmium (Cd) > Nickel (Ni) > Cobalt (Co) > Manganese (Mn) > Zinc (Zn). Among these heavy metals, the concentration of Chromium (Cr) was found to be highest with an average value of 1.320 mg/L. The industrial area of Kasur is reported to be the major source of most of these toxic heavy metals [41]. In a study conducted in Sialkot, Pakistan, the concentrations of various heavy metals in the groundwater were found to be higher than the acceptable values specified by World Health Organization (WHO) for these parameters. These included concentrations of Nickel (Ni) at 0.10 mg/L, Copper (Cu) at 0.06 mg/L, Lead (Pb) at 0.49 mg/L, Manganese (Mn) at 0.03 mg/L, Iron (Fe) at 0.30 mg/L, Chromium (Cr) at 0.03 mg/L and Zinc (Zn) at 0.16 mg/L. The area under study was divided into six regions. Based on the results, it was observed that the extent and causes of groundwater pollution were different in all the regions of the study area. In another study, the concentrations of lead in the untreated and treated surface water samples collected from 18 districts of Karachi were found to be within the permissible limits. The mean concentrations of lead in untreated and treated surface water samples were found to be higher in groundwater than in surface water (Haq et al. 2011).

This work found elevated levels of trace elements similar to a number of earlier surveys conducted in Pakistan. All of these studies reported widespread contamination of groundwater by human activities. Trace elements are among the more persistent environmental contaminants and can have toxicity for ecological organisms and human health. However, impact can vary greatly when more than one element is present at the same time [18, 39, 54]. Water from tubewells in Kasur industrial area contained levels of Cr, Fe, Ni, Cd, Pb, Zn, Co and Mn that were higher than the WHO recommended levels for drinking water by a factor of four to twenty-six. Industrial activities were the principal cause of these high levels [41]. Groundwater from Sialkot contained high levels of Ni, Cu, Pb, Mn, Fe, Cr and Zn that were above the WHO recommended levels for drinking water. The levels of these trace elements in groundwater from different parts of Sialkot were found to vary greatly (Ul Haq et al. 2011). Seasonal variation in levels of trace elements in groundwater was reported in another study. In that work Pb, Zn and Cu were found to be the most persistent contaminating the groundwater during winter, while Ni and Cr were found to be the most persistent during summer; thus, hydrogeological factors as well as season could affect the distribution of these contaminants in groundwater [26]. Recently, it was reported that the levels of several trace elements in tube-well water from southern Punjab were at levels that could pose serious risks to human health [47]. A study that used water samples from rural tube-wells in Tehsil Jhang found that 67% of the water samples from these locations contained levels of trace elements that exceeded the WHO recommended limits for these contaminants in drinking water [31, 33].

The significantly higher concentrations of Ni, Pb, and Cu observed in industrial water indicate that anthropogenic activities are the dominant contributors to trace element contamination in the study area. Bahawalpur has experienced increasing industrialization, including textile processing, metal fabrication, agro-based industries, fertilizer production, engineering workshops, and municipal wastewater discharge. Effluents generated from these activities may contain elevated concentrations of trace

elements that are discharged directly or indirectly into nearby drains and water bodies. The elevated concentrations observed in industrial water therefore reflect continuous inputs from untreated or partially treated industrial effluents, whereas comparatively lower concentrations in surface and groundwater suggest dilution, natural attenuation, and geochemical filtration processes. Ni and Cu levels in the different water samples of the investigated area were found to be above the WHO recommended values for drinking water, for most of the samples (Fig. 2 and Table 2). These levels of contamination would mostly be due to increased industrial activities of industrialization, increased agricultural activities, the presence of untreated wastewater, and others that are known to cause the pollution of trace elements in Pakistan [2, 36, 57].

Jehan et al. [27] reported elevated levels of Pb and Cu in water samples from industrial estates of Pakistan. Concentrations of trace elements in industrial water are substantially higher than in surface water and groundwater. Industrial effluents are mostly untreated or are being treated inadequately, leading to the release of contaminated water into the environment. In aquatic systems, trace elements can accumulate in sediments and persist for long periods of time. These elements can be taken up by benthic organisms and get transferred to the upper food chain through aquatic food webs, posing ecological as well as human health risks [2, 22]. High levels of trace elements in sediments of Nullah Aik have been reported by Qadir et al. [45]. In the midstream section of the Nullah, the highest levels of trace elements were found in industrial effluents. In the present study, it was observed that industrial water contained higher levels of Ni, Pb, and Cu than groundwater and surface water, indicating that industrial activities are a major source of trace elements in the study area.

Similar contamination patterns have been reported in mining and industrial regions worldwide, where elevated concentrations of Ni, Pb, and Cu have been linked

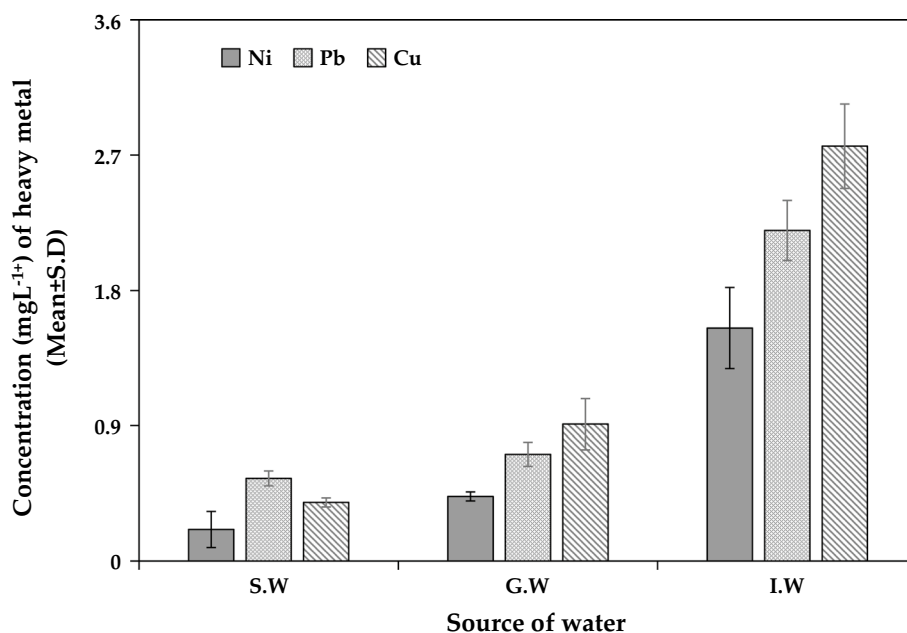


Fig. 2 Mean concentration (mg/L) of trace elements (Ni, Pb, and Cu) in various sources of water (S.W, G.W, and I.W) from the district of Bahawalpur, Punjab, Pakistan

to mining operations, ore processing, smelting activities, waste-rock weathering, and acid mine drainage. Although Bahawalpur is not a major mining district, the contamination observed in the present study demonstrates that industrial discharges, agricultural runoff, and municipal wastewater can generate comparable environmental pressures on water resources. These findings indicate that different anthropogenic activities may produce similar trace element contamination pathways despite differences in regional geological settings [35, 44].

Elevated concentrations of trace elements may adversely affect aquatic ecosystems through their accumulation in sediments and subsequent transfer into aquatic food webs. Because trace elements are persistent and non-biodegradable, they may bioaccumulate in aquatic organisms and biomagnify through higher trophic levels, thereby impairing growth, reproduction, physiological functions, and overall ecosystem stability. Consequently, industrial wastewater not only represents a potential risk to human health but also threatens freshwater biodiversity and ecological integrity [9, 22].

This study demonstrated that the trace elements in ground and surface water have a direct impact on human consumption, while the impact of industrial water is indirectly effective. This is because the surface and ground water are directly used by the human population. Industrial effluents can enter receiving surface-water systems, contaminate sediments, and potentially contribute to groundwater contamination through infiltration, thereby creating indirect pathways for ecological and human exposure. Trace elements released from industrial and other anthropogenic sources may also accumulate in aquatic organisms and move through aquatic food webs, potentially increasing ecological and human health risks [12, 20]. In coherence with the current study, the target hazard quotient (THQ_{oral}) for Pb in drinking water from the groundwater was recorded to be higher, indicating a significant risk as reported by Boumaza et al. [8] and Naz et al. [41].

The elevated THQ values observed for Pb in groundwater and for all three investigated trace elements in industrial water indicate that prolonged exposure may pose unacceptable non-carcinogenic health risks, particularly through oral consumption of contaminated water. Children are especially vulnerable because of their higher water intake relative to body weight and their developing nervous and immune systems. Although carcinogenic risk values remained below internationally accepted thresholds, continuous exposure to trace elements may increase cumulative health burdens over time, emphasizing the need for regular monitoring, effective water treatment, and preventive environmental management [30].

Overall, the findings indicate that anthropogenic activities, particularly industrial discharges and intensive agricultural practices, are the principal drivers of trace element contamination in the investigated water sources. The observed differences among industrial, groundwater, and surface water reflect variations in contaminant inputs, hydrogeological transport processes, and natural attenuation mechanisms. These findings emphasize the need for routine monitoring, effective wastewater treatment, stricter industrial discharge regulations, and sustainable agricultural management to protect drinking water resources and minimize long-term ecological and human health risks.

5 Conclusion

This study evaluated the concentration of nickel (Ni), lead (Pb), and copper (Cu) in surface water, groundwater, and industrial water collected from 6th tehsils of District Bahawalpur, Punjab, Pakistan, and assessed the associated human health risks. The results demonstrated significant differences among the three water sources, with industrial water exhibiting the highest trace element concentrations, followed by groundwater and surface water. Although dermal exposure and carcinogenic risk remained within acceptable limits, the THQ and TTHQ values for oral exposure exceeded the recommended safety threshold in several water sources, indicating potential non-carcinogenic health risks associated with the long-term consumption of untreated water. These findings emphasize the importance of continuous monitoring and effective management of drinking water quality in the study area.

The findings provide valuable baseline information for regional water quality management. Policymakers and environmental authorities should strengthen and routine water quality surveillance, enforce stricter regulations on industrial effluent discharge, and implement integrated pollution control strategies to minimize trace element contamination from industrial, agricultural, and domestic sources. Water treatment agencies should prioritize appropriate treatment technologies and regular monitoring to ensure compliance with the national and international drinking water quality standards, while public awareness programs should be promoted to reduce human exposure to contaminated water.

The present study provides important baseline information on the occurrence of Ni, Pb, and Cu and their associated human health risk in the water sources investigated of district Bahawalpur. Future studies could further strengthen these findings by incorporating seasonal monitoring, a broader range of potentially toxic elements and microbial contaminants, child-specific exposure assessment, hydrogeochemical, and source apportionment analyses, and advanced health risk assessment approaches to provide a more comprehensive understanding of water contamination and support evidence-based water resource management in Bahawalpur and other regions facing similar environmental challenge.

Supplementary Information

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Supplementary file 1.

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Author contributions

S.N.: Writing—original draft, formal analysis, data curation, and review and editing. A.M.M.C. and Q.U.: Conceptualization, writing—review and editing, and supervision. M.F.: Review and editing, methodology, investigation. H.J.: Resources, software, investigation. R.H.: Resources and visualization. S.N.: Drafting, supervision, writing—review and editing.

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Data availability

All relevant data are included within the article.

Declarations

Ethics approval and consent to participate

Ethical approval of this study to use animals as research subjects, including all the procedures, was obtained from the Institutional Animals Ethics Committee (IAEC) of Government Sadiq College and Women University, Bahawalpur. The guidelines issued by the Institutional Animals Ethics Committee were strictly followed. Informed consent was obtained from all individual participants included in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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